

CAGED: Calibrated Agentic Guideline-grounded Early-exit Diagnostics for Multi-Domain Medical Reasoning

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1. BACKGROUND AND MOTIVATION

Every medical agentic system currently faces a dilemma between being thorough enough to be accurate whilst also being efficient enough to be deployable. The deeper you go by invoking more tools and gathering more measurements, the better your diagnosis tends to get but running every tool on every case is expensive, slow, and in a real clinical environment, often unnecessary which makes finding the right trade-off point really important for applying these frameworks. For example, a clear-cut glaucoma presentation does not need a VLM analysis if the CDR and IOP already agree, so, the main question is that "when has the system seen enough to commit?"

PASS is currently the state-of-the-art framework which answers this by training a probabilistic controller to learn which tools to invoke and when to stop. It works well, and the results on the CAB-E benchmark developed by them are also strong but the confidence signal lives inside the model weights so there is no explicit or readable posterior over diagnostic category that a clinician can inspect about and because the controller is trained on chest X-ray tool graphs with labeled reasoning trajectories, moving it to a different domain means rebuilding and retraining from scratch altogether.

My previous report on AdaptiMed pointed in the same direction but with simpler approach. A CV gate is applied to accumulated confidence, with fixed thresholds tuned to the EGS 2020 glaucoma guideline. On 50 REFUGE2 cases, that improved broad accuracy from 54% to 80% and reduced average tool calls by 26%. The gain came from letting the system exit early when evidence was already sufficient instead of always running every tool that is available.

That result validates the hypothesis but the heuristic is brittle. The thresholds ($\theta_1 = 0.72$, $\delta = 0.30$) were hand-tuned for glaucoma under one specific guideline. They do not transfer to heart failure staging under ACC/AHA or to rare diseases without manual recalibration. And the majority vote mechanism cannot distinguish genuine diagnostic uncertainty from LLM temperature noise. More realistically, AdaptiMed is a single-domain system and it was built for glaucoma and it only works for glaucoma.

FETALFusion, which I have submitted to MICCAI 2026 addresses a specific problem, i.e, models trained on one acquisition protocol often fail on another. The solution was a routing bottleneck that learns two sets of channel gates during training, one per protocol, and then simply averages them at inference so no protocol label is needed at deployment. The practical result is a model

that generalises across the acquisition protocols without being retrained or reconfigured for each one and CAGED applies the same inference-time principle to clinical reasoning, which is, instead of a learned gate, a guideline retrieval step makes the per-case decision, and instead of protocol-specific training data, it uses dynamic context from the retrieved guideline.

This proposal pursues the goal, which is, a "single trained agentic diagnostic system" that transfers to an unseen clinical domain at inference time using only a retrieved guideline, with no retraining, no domain labels, and no curated domain-specific trajectory data.

2. PROBLEM STATEMENT

Consider a patient case represented by a medical image, a clinical query, and some patient context. An agentic system works through this case in tiers where each tier runs a set of tools and returns structured evidence. After each tier, the system faces a choice, which is, "is what I know now enough to diagnose, or do I need to go deeper?"

Suppose, let $E_{1:k}$ be the evidence accumulated through tier k , and let $P(Y | E_{1:k})$ be the posterior over diagnostic categories. The system should exit at tier k if and only if:

$$H(P(Y | E_{1:k})) \leq \tau_k^G \wedge \mathbb{E}[\Delta \text{KL}_{k+1}] \leq \epsilon \quad (1)$$

The first condition says the posterior is concentrated enough meaning that the system is not genuinely confused about the diagnosis. The second says there is little expected gain from going deeper, so the next tier is unlikely to change the answer materially. The threshold τ_k^G is not a fixed scalar; it is derived from the clinical guideline relevant to this patient's presenting complaint. What counts as sufficient evidence is defined by the guideline and not by the training data.

Three things that make this different from prior work are, that:

- i) It is **designed/principled** to be grounded in Bayesian evidence accumulation rather than hand-tuned scalars
- ii) It is **guideline-grounded** as the exit criterion is traceable to a named clinical standard so a clinician can see what evidence threshold was applied and why it was applied.
- iii) It is **transferable** as the posterior update mechanism is disease-agnostic, so the same framework works across ophthalmology, cardiology, and rare disease without retraining needed.

3. LIMITATIONS OF CURRENT METHODS

PASS deals with adaptive agentic depth. I found 3 limitations I would like to address, which are :

- i) Its controller requires contrastive path-ranking data to train labeled pairs of good and bad reasoning trajectories that are expensive to curate for each new domain. Moving to cardiology means new data, a new supernet, and a new round of cost-aware RL.
- ii) The confidence signal is implicit in learned weights. There is no inspectable posterior, so a clinician can't audit why the system decided to exit at a given tier.
- iii) Exit decisions are learned from data distributions and not grounded in what clinical guidelines define as evidentially sufficient. That gap matters when the system is expected to operate in regulated clinical environments.

None of the frameworks I have studied report any calibration metrics. This is a critical gap as any system whose early-exit decision depends on a confidence signal must first demonstrate that the signal is trustworthy. An uncalibrated posterior makes the exit criterion in Eq. 1 meaningless regardless of accuracy.

AdaptiMed sidesteps the training cost but trades it for brittleness. The thresholds are hand-tuned for one disease under one guideline. The majority vote conflates stochastic LLM noise with real diagnostic uncertainty. The ablation could not isolate the CV gate's independent contribution as that requires a full rerun. And the EvidenceTierPrioritiser, as correctly noted in the feedback by Dr.Chen, is a static lookup table with no conditional branching logic. It cannot handle the multi-criteria staging that appears in guidelines like ACC/AHA heart failure or rare disease protocols.

The limitation is shared across all surveyed frameworks on agentic ai in medical diagnosis which is, that every system is developed and evaluated on a single domain, with no demonstration of cross-domain transfer without retraining. That is the deployment barrier this work will directly target.

4. RESEARCH QUESTIONS

Some of the questions I'd like to work towards are :

- Q1. The main idea behind AdaptiMed was that a CV gate on accumulated confidence would know when to stop. It worked, but the thresholds were adjusted by hand and the confidence signal was never proved to be reliable. Can a Bayesian posterior updated incrementally across tiers produce superior exit decisions, as assessed by ECE and diagnostic accuracy, in comparison to a fixed-threshold approach?
- Q2. Can per-case guideline retrieval set entropy thresholds that can be traced back to a particular clinical criteria? And does this automatically handle the conditional branching logic that a static schema like the EvidenceTierPrioritiser can't?
- Q3. Does CAGED transfer to a new and unseen clinical

domain at inference time without retraining? If yes, then how does it compare to PASS on CAB-E, where PASS was built and tuned?

5. PROPOSED METHODOLOGY

5.1. Component A: Bayesian Diagnostic Posterior

After each tier, CAGED updates a probability distribution over diagnostic categories using a Dirichlet posterior. Think of it as a running tally, where each tool's output shifts the distribution toward whichever diagnosis its evidence best supports, and the guideline determines how much weight each tool's contribution carries. A tool the guideline treats as primary, like IOP in glaucoma, shifts the distribution more than a supplementary one. The result is a probability estimate that is fully inspectable at any point, so a clinician can see exactly which measurements moved the needle and by how much. One assumption is that the update treats each tool's output as independent of the others given the diagnosis. For correlated measurements like CDR and RNFL this is likely violated, and relaxing it is an open modelling question for later work. This replaces AdaptiMed's majority vote, which treated three LLM calls as independent opinions, with an accumulation mechanism that treats tool outputs as evidence that compounds across tiers.

5.2. Component B: Guideline-RAG Threshold Retrieval

Before tier 1, a lightweight RAG query retrieves the relevant section of the applicable clinical guideline like EGS 2020, ACC/AHA, etc for the patient's presenting complaint. Two things will be extracted, which are, (i) a per-tier sufficiency criterion τ_k^G , the entropy level at which the accumulated evidence is guideline sufficient for a diagnosis and (ii) a per tool reliability prior λ_j , encoding which measurements the guideline treats as primary vs supplementary.

This is the principled version of what I was attempting with the EvidenceTierPrioritiser; since it retrieves dynamically at inference time and it handles conditional branching naturally, the retrieved section for a specific case carries different staging logic than a disease of different domain, and both are handled without manual schema modification. So, no retraining is required when switching domains.

5.3. Component C: Expected Information Gain Estimation

Before calling the next tier, CAGED asks a simple question, "is the next tier actually likely to change the answer?" To estimate this, it simulates what the probability distribution would look like after the next tier runs using the reliability weights and measurement priors already retrieved from the guideline. If the simulated update is small enough, meaning if the distribution barely moves,

the system exits. If it would shift the distribution above a specific threshold, the system will go deeper. As this estimate comes directly from the same Dirichlet framework and guideline priors that Components A and B already maintain, Component C needs no separate model and no domain-specific training data. So, every decision, whether to continue or to stop, is logged alongside the full posterior state. This gives a justification or reasoning behind what the model did and why.

5.4. Evaluation Plan

Baselines will be MedAgent-Pro, AdaptiMed, and PASS (reported numbers on overlapping benchmarks). Datasets I plan to use are REFUGE2 glaucoma using the existing AdaptiMed setup with real CDR from segmentation masks, CAB-E for CXR for enabling direct comparison with PASS on its home benchmark; and one held-out cardiology dataset to test Q3, subject to labs guidance and data access. Metrics will include category accuracy, broad accuracy, average tool calls, early-exit rate, and expected calibration error which is, by what I observed, absent from all five surveyed frameworks yet essential and I want to use it because CAGED's exit criterion in Eq. 1 is only valid if the Dirichlet posterior is well-calibrated. Therefore, ECE is not an auxiliary metric but the primary validation of the core mechanism and an essential metric we can use to prove the claim for any clinical deployment argument. The ablation follows the same four-configuration structure as AdaptiMed to isolate each component independently, and includes the independent Config C rerun that the original report could not perform.

I'm also open to the lab's suggestions on datasets, the scope of evaluation, any section of the methodology, or other ways that the lab's experience might improve the approach.